Fixed-Resource Undecidability of Homogeneous Polynomial Global Asymptotic Stability

A detailed mathematical proof and guide to the complete Lean formalization

Formalization project companion

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Abstract

For fixed natural numbers n, d, let $\mathrm{HGAS}_{\mathbb{Q}}(n, d)$ be the set of finite sparse rational codes for homogeneous polynomial vector fields $F : \mathbb{R}^n \to \mathbb{R}^n$ of degree d whose origin is globally asymptotically stable. Invalid homogeneity codes are declared negative instances.

This document proves that there are fixed constants $n_* \geq 1$ and odd $d_* \geq 3$ for which $\mathrm{HGAS}_{\mathbb{Q}}(n_*, d_*)$ is Σ_1^0-complete under computable many-one reductions. In particular, the fixed-pair decision problem is undecidable and its complement is not computably enumerable. More strongly, there is a fixed threshold N_* such that

$$\mathrm{HGAS}_{\mathbb{Q}}(N, d) \quad \text{is } \Sigma_1^0\text{-complete whenever} \quad N \geq N_*, \quad d \geq 3, \quad d \text{ odd}.$$

Thus cubic homogeneous GAS is already Σ_1^0-complete in every fixed dimension $N \geq N_*$.

The lower bound compiles universal halting into a fixed diagonal–quadratic– linear integer system, a prime-radical torus detector, a polynomial modular/Jacobi source, a compact stereographic accumulator, and finally a rational homogeneous polynomial field. The upper bound enumerates exact rational Lyapunov certificates. A solvability-preserving augmentation $x_j - y_j = 0$ is essential: it forces every detector column to have a nonzero linear character and thereby removes a genuine zero-ratio resonance in an earlier three-moment argument.

Every theorem used in the final dependency chain is formalized in the accompanying Lean project. The final theorems contain no `sorry`, `admit`, project `axiom`, or `unsafe` declaration. Kernel inspection of the headline results reports only `propext`, `Classical.choice`, and `Quot.sound`. The fixed constants are named finite objects selected once and for all; because the particular MRDP representative is obtained by classical choice, the development does not claim that n_*, d_*, N_* have been evaluated to decimal numerals.

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1 The theorem, exactly stated

1.1 Finite syntax and semantics

A *polynomial field code* in dimension n is an n-tuple of finite lists of pairs

$$(\alpha, c), \qquad \alpha \in \mathbb{N}^n, \quad c \in \mathbb{Q}.$$

Its realization is the rational polynomial vector field

$$F_i(x) = \sum_{(\alpha,c) \text{ in coordinate } i} c\, x^{\alpha}.$$

Duplicate monomials and zero coefficients are allowed in the raw syntax; decoding performs the usual finite polynomial sum. There is a primitive recursive Boolean test saying that every decoded coordinate is homogeneous of degree d.

The formal predicate is

$$\begin{aligned} \mathrm{HGAS}_{\mathbb{Q}}(n,d)(F) \quad \Longleftrightarrow \quad & F \text{ passes the degree-}d \text{ homogeneity test,} \\ & 0 \text{ is globally asymptotically stable for } \dot{x} = F(x). \end{aligned} \tag{1.1}$$

This is `HGASQ` in `HGAS/Computability/HGASLanguage.lean`. Global asymptotic stability is not defined through a possibly nonexistent global flow. It contains four separate assertions:

1. the origin is an equilibrium;
2. every initial point admits a forward solution on $[0, \infty)$;
3. the origin is Lyapunov stable; and
4. every forward solution converges to the origin.

These definitions occur in `HGAS/Dynamics/Stability.lean`. This explicit forward-completeness clause matters because polynomial ODEs may otherwise blow up in finite time.

Definition 1.1. *A predicate P on a primitively codable type is Σ_1^0-complete if P is computably enumerable and every computably enumerable predicate on $\mathbb{N}$ computably many-one reduces to P.*

The project uses Mathlib's partial-recursive codes and proves this definition equivalent to reduction from its universal halting predicate; see `HGAS/Computability/SigmaOne.lean`.

1.2 The fixed constants

The constants used in the formal statement are deliberately defined from the actual compiler, rather than hidden behind an existential quantifier. Let E_0, V_0 be the fixed row and variable counts of the selected universal DQL representation. The repair in Section 3.4 uses

$$E_1 = E_0 + V_0, \qquad V_1 = 2V_0. \tag{1.2}$$

Let n_{src} be the resulting compact-source dimension and let D_{wrap} be its fixed odd wrapper degree. The homogeneous product-sphere wrapper uses two source sphere blocks and adds one projective coordinate, so define

$$n_* = n_{\mathrm{src}} + 1, \qquad d_* = D_{\mathrm{wrap}} + 2(2+1) = D_{\mathrm{wrap}} + 6. \tag{1.3}$$

The literal Lean definitions are `fixedDimension` and `fixedDegree` in `HGAS/Arithmetic/NormalizedCorrectedAugmentedEffectiveFamily.lean`. The same file proves $n_* > 0$, d_* odd, and $d_* \geq 3$ as `fixedDimension_pos`, `fixedDegree_odd`, and `fixedDegree_ge_three`.

Put

$$k_* = \frac{d_* - 1}{2}, \qquad N_* = \#\{\alpha \in \mathbb{N}^{n_*} : |\alpha| = k_*\}. \tag{1.4}$$

The last cardinality is the dimension of the degree-k_* Veronese space. In LEAN it is

`Veronese.cubicLiftCodeDimension(fixedDimension, fixedOutputHalfDegree)`.

Theorem 1.2 (Fixed-pair and cofinal completeness)**.** *With the fixed constants above:*

1. $\mathrm{HGAS}_{\mathbb{Q}}(n_*, d_*)$ *is* Σ_1^0*-complete;*
2. *for every* $N \geq N_*$ *and every odd* $d \geq 3$, $\mathrm{HGAS}_{\mathbb{Q}}(N, d)$ *is* Σ_1^0*-complete; and*
3. *in particular, for every* $N \geq N_*$, $\mathrm{HGAS}_{\mathbb{Q}}(N, 3)$ *is* Σ_1^0*-complete.*

The three kernel-checked declarations are, respectively,

`fixedPair_sigmaOneComplete`,
`cofinal_sigmaOneComplete`,
`cubic_sigmaOneComplete`

in `HGAS/Arithmetic/NormalizedCorrectedAugmentedEffectiveFamily.lean`. The same file records the undecidability consequences as `fixedPair_not_computable` and `cubic_not_computable`, and the non-c.e.-complement consequences as `fixedPair_complement_not_re` and `cubic_complement_not_re`.

Remark 1.3 (What is not claimed)**.** *The theorem proves cubic undecidability in every sufficiently large fixed dimension. It does* not *decide the separate low-dimensional problem* $\mathrm{HGAS}_{\mathbb{Q}}(3, 3)$*. Nothing in the Veronese or padding argument lowers the fixed dimension to three.*

Remark 1.4 (Fixed does not mean numerically printed)**.** *The universal Diophantine representation is a finite object chosen from a proved existence theorem. Once chosen, its dimensions and coefficient tables are constants, and all input-specialization maps are proved computable. But the choice is noncomputable in the programming-language sense, so the formal development does not advertise a command* `#eval` *that prints* n_**. The theorem is an ordinary fixed-pair existence and many-one completeness theorem, not a numerical resource-minimization result.*

1.3 Proof map

The proof has two branches:

$$\begin{gathered}\text{homogeneous GAS} \Longrightarrow \text{finite exact rational Lyapunov certificate} \Longrightarrow \mathrm{HGAS}_{\mathbb{Q}}(n, d) \in \Sigma_1^0, \\ \text{universal halting} \Longrightarrow \text{fixed augmented DQL} \Longrightarrow \text{compact source with a strict invariant-average sign} \\ \Longrightarrow \text{fixed homogeneous rational field} \Longrightarrow \text{cubicization and cofinal padding.}\end{gathered} \tag{1.5}$$

The remainder of the document proves each arrow and identifies its LEAN counterpart.

2 The computably enumerable upper bound

Theorem 2.1 (Uniform c.e. upper bound at fixed odd degree)**.** *For every fixed n and odd d, the predicate* $\mathrm{HGAS}_{\mathbb{Q}}(n,d)$ *is computably enumerable.*

This is `hgasQ_re_of_oddDegree` in `HGAS/Lyapunov/RationalHomogeneousConverse.lean`. We give the full mathematical argument because the lower bound alone would establish hardness, not Σ_1^0-completeness. Its formal scope is one fixed dimension and one fixed odd degree. The manuscript's coded variable-resource and even-degree upper bound instead uses a paper-level real quantifier-elimination argument and is not a separate final Lean declaration.

2.1 The exact certificate

Write

$$\rho(x) = x_1^2 + \cdots + x_n^2.$$

A certificate contains a rational polynomial p, a natural exponent r, and two natural grid resolutions m_p, m_q. The checker generates the finite rational grids, coefficient–derivative bounds, and rational margins from these data. The certificate represents the degree-two homogeneous rational function

$$V(x) = \frac{p(x)}{\rho(x)^r}. \tag{2.1}$$

If F has degree d, differentiation gives

$$\dot{V}(x) = \frac{\rho(x)\nabla p(x)\cdot F(x) - 2r\,p(x)\,x\cdot F(x)}{\rho(x)^{r+1}}.$$

Thus the polynomial numerator used to certify strict decrease is

$$q(F,p,r) = -\rho\,\nabla p\cdot F + 2r\,p\,x\cdot F. \tag{2.2}$$

The semantic polynomial is `HGAS.rationalLyapunovNumerator` in `HGAS/Lyapunov/RationalCertificateAlgebra.lean`. Its executable sparse implementation is `PolynomialCode.rationalLyapunovNumerator` in `HGAS/Foundation/PolynomialDifferentiationCode.lean`; a decoding theorem identifies the two exactly.

Although ρ in (2.1) is the squared Euclidean norm, homogeneity makes strict positivity of p and q away from zero equivalent to strict positivity on the unit sphere of any norm. The Boolean checker uses the sup-norm unit sphere, whose boundary admits a particularly simple finite rational grid. Exact derivative bounds transfer the checked grid margin to every point of that sphere. The finite checker and its semantic interpretation are developed in `HGAS/Lyapunov/ExactGridPositivity.lean` and `HGAS/Lyapunov/GridRationalCertificateSoundness.lean`; primitive recursiveness is proved in `HGAS/Lyapunov/ExactGridPositivityComputability.lean` and `HGAS/Lyapunov/GridRationalCertificate.lean`.

If it accepts, (2.1) is positive definite away from zero and (2.2) is positive away from zero. Coercivity of V bounds forward trajectories inside compact sublevel sets. Standard local ODE continuation then proves forward completeness; the strict derivative inequality gives Lyapunov stability and global attraction. Hence accepted certificates are sound for the exact GAS definition, including completeness. This is proved in `HGAS/Lyapunov/GridRationalCertificateSoundness.lean`.

2.2 Completeness of the certificate search

Suppose now that F is homogeneous of odd degree d and is GAS. The completeness proof has five steps.

1. A homogeneous C^1 Lyapunov function. Separate radius and direction. On the unit sphere put

$$a(u) = \langle u, F(u) \rangle, \qquad Y(u) = F(u) - a(u)u. \tag{2.3}$$

Then Y is tangent to the sphere. For $x = ru$, homogeneity gives, after the usual angular time change,

$$\frac{\mathrm{d}u}{\mathrm{d}s} = Y(u), \qquad \frac{\mathrm{d}}{\mathrm{d}s} \log r = a(u). \tag{2.4}$$

GAS yields a uniform negative radial-cocycle block on the compact sphere. Integrating a suitably smoothed finite-horizon cocycle produces a strict degree-two homogeneous C^1 Lyapunov function. The construction is formalized in `HGAS/Lyapunov/HomogeneousConverseC1.lean` and `HGAS/Lyapunov/EuclideanAngularConverse.lean`. It is closely related to Rosier's homogeneous converse theorem [21], but the particular theorem needed here is proved internally.

2. Evenization. Odd degree implies $F(-x) = -F(x)$. If V is a strict Lyapunov function, then

$$V_{\mathrm{ev}}(x) = \frac{V(x) + V(-x)}{2} \tag{2.5}$$

is even, positive, degree two, and has strictly negative orbital derivative. The proof, including the derivative identity, is reusable in `HGAS/Analysis/EvenC1Lyapunov.lean`.

3. Simultaneous C^1 polynomial approximation. On a cube containing the unit sphere, tensor Bernstein polynomials approximate V_{ev} and all first partial derivatives uniformly. The development proves the tensor derivative formula rather than assuming a black-box approximation theorem; see `HGAS/Analysis/TensorBernsteinPolynomial.lean`, `HGAS/Analysis/TensorBernsteinDerivative.lean`, and `HGAS/Analysis/TensorBernsteinC1Approximation.lean`. For sufficiently small C^1 error, positivity and strict decay retain positive margins on the sphere.

4. Homogenization and rational perturbation. Evenness permits reconstruction of an even homogeneous numerator after multiplication by a sufficiently high power of ρ. Rational coefficients are dense in the finite coefficient space, so a sufficiently small rational perturbation preserves both compact strict margins. This produces a rational function of the form (2.1).

5. A finite exact grid witness. Strict positivity on a compact sphere supplies positive margins and finite derivative bounds. A sufficiently fine rational grid therefore passes the exact checker. This is the completeness theorem in `HGAS/Lyapunov/GridRationalCertificateCompleteness.lean`.

Consequently every positive instance has some finite accepted certificate. Enumerating all finite candidate codes semidecides GAS. This rational certificate viewpoint is consistent with the algebraic Lyapunov results of Ahmadi and El Khadir [2]; here, however, the exact certificate language, soundness, approximation, and enumerability chain are all present in the kernel-checked development.

3 The universal arithmetic source

3.1 Universal halting

MATHLIB represents partial-recursive functions by a countable type of codes. Define

$$\text{UniversalHalts}(e, x) \quad \Longleftrightarrow \quad \text{eval}(e, x) \text{ is defined.} \tag{3.1}$$

Its domain is c.e. Conversely, if $A \subseteq \mathbb{N}$ is c.e., choose a code e_A for the partial computation which returns 0 precisely on members of A. Then

$$x \in A \quad \Longleftrightarrow \quad \text{UniversalHalts}(e_A, x). \tag{3.2}$$

The map $x \mapsto (e_A, x)$ is computable. This proves universal-halting Σ_1^0-completeness in `HGAS/Computability/SigmaOne.lean`.

For the arithmetic compiler it is convenient to encode the code–input pair by one natural number. The predicate `UniversalHalting.Predicate` : $\mathbb{N} \to \text{Prop}$ is the resulting unary complete predicate. Its exact encoding and bounded interpreter are in `HGAS/Arithmetic/ExecutableMRDPBoundary.lean`.

3.2 The formal MRDP fragment

The reduction needs a fixed polynomial $P(t, z_1, \ldots, z_s) \in \mathbb{Z}[t, z_1, \ldots, z_s]$ such that

$$\text{Predicate}(t) \quad \Longleftrightarrow \quad \exists z \in \mathbb{N}^s,\ P(t, z) = 0. \tag{3.3}$$

The project does not introduce (3.3) as an axiom. It proves the precise primitive-recursive Diophantine closure statement needed for the bounded universal-machine trace.

Here is the mathematical core. Diophantine graphs are plainly closed under the initial functions and composition. Primitive recursion asks one to express a finite trace

$$a_0 = g(\bar{x}), \qquad a_{i+1} = h(\bar{x}, i, a_i) \quad (0 \le i < N), \qquad a_N = y \tag{3.4}$$

with a fixed number of existentially quantified natural variables. A sufficiently large radix β packs the sequence into one integer

$$A = \sum_{i=0}^{N} a_i \beta^i. \tag{3.5}$$

Sparse radix ciphers provide Diophantine access to the digits and to finite aligned register operations. The conditions in (3.4), apparently a bounded universal quantification over $i < N$, are combined into a single sparse bounded-forall certificate. The radix is itself certified large enough that carries cannot corrupt any register. Correctness has two directions:

- a genuine finite computation produces the packed certificate; and
- every accepted packed certificate decodes, digit by digit, to a genuine computation satisfying (3.4).

The principal modules are `HGAS/Arithmetic/SparseRadixCipher.lean`, `HGAS/Arithmetic/SparseCipherCircuit.lean`, `HGAS/Arithmetic/SparseBoundedForall.lean`, and `HGAS/Arithmetic/PrimitiveRecursiveDiophantine.lean`. Their endpoints include `cipherCertificate_iff_boundedForall`, `boundedForall_dioph`, `recursionCertificate_iff`, `natPrimrec_unaryGraph_dioph`, and `binaryPrimrecRel_dioph`.

Applying the last theorem to the primitive-recursive bounded interpreter proves

`binaryPrimitiveRecursiveRelationsAreDiophantine`

in `HGAS/Arithmetic/UnconditionalMRDP.lean`. Finite-support extraction then supplies (3.3). This is the portion of the classical Davis–Putnam–Robinson–Matiyasevich theorem [10, 16, 17, 18] needed by the reduction, formalized rather than postulated.

3.3 Fixed diagonal–quadratic–linear systems

A diagonal–quadratic–linear, or DQL, system with E rows and V integer variables is

$$\sum_{j=1}^{V}(A_{ij}x_j^2 + B_{ij}x_j) = c_i, \qquad i = 1, \dots, E, \tag{3.6}$$

with all coefficients integral. Mixed quadratic terms are forbidden. This restriction loses no effectiveness. A multiplication gate $xy = z$ can be replaced by a fresh variable w and

$$w = x + y, \qquad w^2 - x^2 - y^2 - 2z = 0. \tag{3.7}$$

Natural-number witnesses are represented over $\mathbb{Z}$ with four squares, using Lagrange's theorem [15]. Addition, constants, copies, and (3.7) compile a finite polynomial circuit into finitely many DQL rows. The equivalence of witnesses is proved in `HGAS/Arithmetic/DQL.lean`, `HGAS/Arithmetic/GateDQLCompiler.lean`, and `HGAS/Arithmetic/FixedRepresentationDQL.lean`.

After the polynomial in (3.3) has been fixed, substituting the input t changes only right-hand-side data in a fixed template. Thus there are fixed E_0, V_0 and a computable map

$$t \longmapsto I_t \in \mathrm{DQL}(E_0, V_0) \tag{3.8}$$

such that

$$I_t \text{ is solvable over } \mathbb{Z} \quad \Longleftrightarrow \quad \mathrm{Predicate}(t). \tag{3.9}$$

These are `unconditionalUniversalHaltingFixedDQL`, `unconditionalUniversalHaltingFixedDQL_computable` and `unconditionalUniversalHaltingFixedDQL_solvable_iff` in `HGAS/Arithmetic/UnconditionalFixedDQL.lean`.

3.4 The nonzero-linear-column augmentation

The final three-moment selector needs the linear character of every arithmetic variable to be nonzero. A general DQL representation need not have this property: a variable may occur only through x_j^2. We enforce it without changing solvability.

For every old variable x_j, add a duplicate y_j and the row

$$x_j - y_j = 0. \tag{3.10}$$

Old rows use only the x-variables. If x solves the old system, (x, x) solves the augmented one. Conversely, the first half of every augmented solution solves all old rows. Therefore

$$\mathrm{Aug}(I) \text{ solvable} \quad \Longleftrightarrow \quad I \text{ solvable}. \tag{3.11}$$

Every old column contains coefficient $+1$ in its new equality row, and every duplicate column contains coefficient -1. Hence every augmented linear column is nonzero.

The construction and both implications in (3.11) are in `HGAS/Arithmetic/DQLNonzeroLinearAugmentation.lean`. The same file proves that augmentation is primitive recursive. Applied to (3.8), it gives fixed dimensions (1.2), a computable family $\widetilde{I}_t$, and

$$\begin{aligned} &\widetilde{I}_t \text{ solvable} \Longleftrightarrow \mathrm{Predicate}(t), \\ &\forall j \ \exists i \quad \widetilde{B}_{ij} \neq 0. \end{aligned} \tag{3.12}$$

The fixed-resource specialization is in `HGAS/Arithmetic/NormalizedCorrectedAugmentedUniversalResources.lean`.

4 Homogeneous GAS as a compact invariant-average sign

The compiler first constructs a compact polynomial source and a continuous reward. This section explains why the sign of the largest invariant average is exactly the right interface to homogeneous GAS.

4.1 Polar dynamics

Suppose F is homogeneous of degree $m+1$, where $m > 0$. For a unit vector u, define a, Y by (2.3). If a nonzero trajectory is written $x(t) = r(t)u(t)$, then

$$\dot{r} = r^{m+1} a(u), \qquad \dot{u} = r^m Y(u). \tag{4.1}$$

With angular time $\mathrm{d}s/\,\mathrm{d}t = r^m$,

$$u' = Y(u), \qquad (\log r)' = a(u). \tag{4.2}$$

Consequently,

$$\log \frac{r(s)}{r(0)} = \int_0^s a(\phi_\tau u(0))\ \mathrm{d}\tau, \tag{4.3}$$

where ϕ is the angular flow on the compact sphere.

4.2 Uniform negative blocks

For a compact flow ϕ and continuous reward a, write

$$\mathcal{M}_\phi(a) = \max_{\mu \in \mathcal{I}(\phi)} \int a\ \mathrm{d}\mu, \tag{4.4}$$

where $\mathcal{I}(\phi)$ is the set of invariant Borel probabilities. The maximum exists. The following equivalence is the key compactness principle:

$$\mathcal{M}_\phi(a) < 0 \quad \Longleftrightarrow \quad \exists T, c > 0 \ \forall x, \quad \int_0^T a(\phi_t x)\ \mathrm{d}t \le -c. \tag{4.5}$$

For the forward implication, if no uniform block exists, choose longer orbit segments with average arbitrarily close to zero. Their occupation probabilities have a weakly convergent subnet. The endpoint error in invariance is $O(1/T)$, so the limit is invariant and has nonnegative reward average, contradicting (4.4). For the reverse implication, integrate the uniform block against any invariant probability.

The occupation-measure argument, existence and compactness of invariant probabilities, and the equivalence (4.5) are formalized in `HGAS/Dynamics/KrylovBogolyubov.lean`, `HGAS/Dynamics/InvariantProbabilityCompact.lean`, and `HGAS/Dynamics/UniformNegativeBlocks.lean`.

Combining (4.2)–(4.5), including the conversion back to physical time, gives

$$0 \text{ is GAS for } F \quad \Longleftrightarrow \quad \mathcal{M}_\phi(a) < 0. \tag{4.6}$$

The reusable formal endpoint is `globallyAsymptoticallyStable_iff_maximalAngularAverage_neg` in `HGAS/Wrapper/AngularAverageGAS.lean`.

4.3 An ergodic maximizer

The compact-accumulator argument needs a maximizing invariant probability which is ergodic for the entire real flow, not merely for its time-one map. The project proves directly that the compact convex set of invariant probabilities has an extreme maximizer for the affine functional $\mu \mapsto \int a\ \mathrm{d}\mu$. It embeds measures into the weak dual of bounded continuous functions, applies Krein–Milman to the maximizing face, and proves that an extreme invariant probability satisfies the full-flow zero–one law by splitting on an invariant measurable set.

The result is `exists_ergodicInvariantProbability_isMaxOn_average` in `HGAS/Dynamics/ErgodicMaximizer.lean`. In particular, to upper bound (4.4), it suffices to upper bound the reward average of every full-flow ergodic invariant probability. No measurable ergodic decomposition theorem is used.

5 The arithmetic Fourier detector

Let $I = (A, B, c)$ be a DQL system with V variables and E rows. Use a torus

$$\mathbb{T}^{V+E} = (\mathbb{R}/\mathbb{Z})^{\{\text{private variables}\}\sqcup\{\text{equation rows}\}}.$$

For $x, s \in \mathbb{Z}^V$, define the private and equation exponents

$$\begin{aligned} p_j(x,s) &= x_j^2 - s_j, \\ q_i(x,s) &= \sum_j (A_{ij} s_j + B_{ij} x_j) - c_i. \end{aligned} \tag{5.1}$$

The full exponent vector is zero exactly when

$$s_j = x_j^2 \text{ for all } j, \qquad I(x) = 0. \tag{5.2}$$

Choose $0 < \lambda, \eta < 1$. The expanded detector is the absolutely convergent Fourier series

$$\mathcal{D}_I(\theta) = \sum_{x,s \in \mathbb{Z}^V} \lambda^{\sum_j |x_j|^2} \eta^{\sum_j |s_j|} e^{2\pi i \langle (p(x,s), q(x,s)), \theta \rangle}. \tag{5.3}$$

Haar integration deletes every nonzero Fourier character. By (5.2),

$$\int_{\mathbb{T}^{V+E}} \mathcal{D}_I \, \mathrm{d}\theta = \sum_{\substack{x \in \mathbb{Z}^V \\ I(x)=0}} (\lambda\eta)^{\sum_j |x_j|^2}. \tag{5.4}$$

Every surviving term is strictly positive. Therefore

$$\int \mathcal{D}_I \, \mathrm{d}\theta \neq 0 \quad \Longleftrightarrow \quad I \text{ is solvable.} \tag{5.5}$$

There is no cancellation argument hidden here: the right side of (5.4) is a nonnegative real series.

The exact support calculation is in `HGAS/Arithmetic/PositiveConstantTerm.lean`; absolute summability, continuity, Haar extraction, and (5.5) are in `HGAS/Arithmetic/DetectorFourier.lean`. The final compiler uses the concrete values

$$\lambda = \frac{1}{256}, \qquad \eta = \frac{1}{2}. \tag{5.6}$$

5.1 Prime-radical clocks

Assign a distinct prime p_ℓ to every fundamental phase coordinate. A polynomial controller on a unit circle has locked equilibria satisfying

$$p_\ell r_\ell^2 = 1, \qquad \omega_\ell = p_\ell r_\ell = \varepsilon_\ell \sqrt{p_\ell}, \qquad \varepsilon_\ell \in \{+1, -1\}. \tag{5.7}$$

The phase circle rotates at angular velocity ω_ℓ. Under $\mathbb{R}/\mathbb{Z} \to S^1,\ \theta \mapsto e^{2\pi i\theta}$, its frequency in turns is $\omega_\ell/(2\pi)$. The detector uses fourth powers of the fundamental phases, so its actual frequency is

$$\Omega_\ell = \frac{2}{\pi} \varepsilon_\ell \sqrt{p_\ell}. \tag{5.8}$$

Square roots of distinct primes are linearly independent over $\mathbb{Q}$. The project proves this by adjoining the roots one at a time and using the square-class obstruction; it does not import the result as an axiom. Signs and the common nonzero scale $2/\pi$ preserve independence. Hence

$$\sum_\ell k_\ell \Omega_\ell = 0, \quad k_\ell \in \mathbb{Z} \quad \Longrightarrow \quad k_\ell = 0 \text{ for all } \ell. \tag{5.9}$$

See `HGAS/Clocks/RadicalIndependence.lean`, `HGAS/Clocks/ScaledRadicalClock.lean`, and `HGAS/Arithmetic/NormalizedCorrectedEffectiveSelectedDetectorPhase.lean`.

For any initial phase and any horizon sequence $T_N \to \infty$, termwise Cesàro integration followed by dominated summation gives

$$\frac{1}{T_N} \int_0^{T_N} \mathcal{D}_I(\theta_0 + t\Omega) \, \mathrm{d}t \longrightarrow \int_{\mathbb{T}^{V+E}} \mathcal{D}_I \, \mathrm{d}\theta. \tag{5.10}$$

The arbitrary-horizon form is important because the recurrent orbit selected later supplies its own physical return horizons. It is proved in `HGAS/Analysis/TorusFourierCesaroArbitraryHorizon.lean` and `HGAS/Arithmetic/DetectorFourierCesaroArbitraryHorizon.lean`.

6 The polynomial modular–Jacobi source

The detector (5.3) is an analytic Fourier series, whereas the reduction must emit a finite polynomial vector field with rational coefficients. The modular–Jacobi source realizes the needed theta factors as coordinates of an autonomous polynomial ODE.

6.1 Unary and Jacobi coordinates

The unary modular block has four complex coordinates

$$(A, C, E, Y).$$

On the canonical theta solution these are, up to the fixed normalization,

$$A = \vartheta_2^4, \qquad C = \vartheta_4^4, \qquad Y = \vartheta_3, \qquad E = E_2. \tag{6.1}$$

Ramanujan/Jacobi differential identities make their derivative under horizontal motion a polynomial in (A, C, E, Y).

The reciprocal Jacobi block has ten complex coordinates

$$(p_2, p_3, p_4, H, T_1, T_2, T_3, T_4, \Delta, U). \tag{6.2}$$

The first four encode theta-square and Eisenstein coefficient data; $T_1, \dots, T_4, \Delta$ encode an elliptic theta state; and

$$U\,T_1 = 1 \tag{6.3}$$

keeps the trajectory in a regular reciprocal chart. There are two commuting rational polynomial derivations: an elliptic derivation and a modular derivation. Their literal sparse Gaussian-rational codes and realification to twenty real coordinates are in `HGAS/Modular/JacobiPolynomialCode.lean`.

The unary and Jacobi blocks are linked, but retained as separate ambient coordinates. Ten polynomial residuals enforce the coefficient identities, the two theta quadrics, the reciprocal chart, and a coherent signed root. These residuals form an invariant ideal. The combined state therefore has fourteen complex, or twenty-eight real, coordinates and an exact finite polynomial compiler; see `HGAS/Modular/LinkedJacobiPolynomialCode.lean`. The formulas are related to the polynomial theta systems studied by Brezhnev, Lyakhovich, and Sharapov [5] and Brezhnev [4]. All normalizations actually used in the reduction are rederived and checked in the project.

6.2 Controllers, phase factors, and residual drifts

There is one linked unary–Jacobi column for each DQL variable. Polynomial controller circles provide the two clock coefficients of each column. Additional real polynomial coordinates provide:

- the fourth-root gauge multiplying T_1;
- two corrected first-moment selectors;
- reciprocal Poisson factors with parameter $1/2$;
- a right-hand-side phase oscillator; and

- nonnegative validity and clock-radius residual observables.

Complex polynomial codes are realified mechanically. The controller and phase circle equations are tangent to their unit circles at every ambient point; the linked residual ideal is preserved; and every displayed ODE identity is proved by evaluation of the finite sparse code. These pieces are assembled in the sequence beginning with `HGAS/Arithmetic/FixedDetectorPolynomialCode.lean` and ending with `HGAS/Arithmetic/NormalizedCorrectedFixedDetectorCompactSourceCode.lean`.

The word "normalized" records three important corrections which are literal in the code:

1. angular velocities are converted from radians to turns by 2π;
2. detector phases use fourth powers, hence the factor 4; and
3. the two first moments are centered at the compiler's exact constant $1/256$.

6.3 From zero conditional drift to a selected orbit

Later, the compact accumulator will imply that every drift coordinate has zero integral under a certain conditional invariant probability. Apply the pointwise ergodic theorem simultaneously to the finite family of drift, speed, and validity coordinates. Poincaré recurrence for the flow and its reverse supplies a point which:

- lies in the finite central chart;
- is recurrent both forward and backward in compact time;
- has zero validity energy, so all circle and linked residual equations hold pointwise;
- realizes the prescribed zero drift means; and
- has a positive average physical-speed coordinate.

The selected object is `SelectedBidirectionalValidityOrbit`. Its construction from conditional zero means is `exists_selectedBidirectionalValidityOrbit_of_conditionalBase_zero` in `HGAS/Arithmetic/NormalizedCorrectedFixedDetectorBidirectionalValidityOrbit.lean`. The bidirectional clause is not cosmetic: standardizing a column by a negative clock frequency reverses time, so forward recurrence alone would not suffice.

7 Classifying a selected Jacobi column

Fix one selected detector column j. Write $\omega_j \neq 0$ for its private modular frequency and N_j for its linear DQL frequency. Rescale time by

$$a_j = \frac{\pi}{\omega_j}. \tag{7.1}$$

The standardized modular clock is exactly πi, while its real elliptic frequency ν_j is defined by

$$\pi\nu_j = a_j \frac{\omega_j + N_j}{2}. \tag{7.2}$$

Both identities, with all signs retained, are proved in `HGAS/Arithmetic/NormalizedCorrectedFixedDetectorColumnStandardization.lean` and `HGAS/Arithmetic/NormalizedCorrectedFixedDetectorCanonicalRootSeed.lean`.

7.1 The level-two horocycle alternative

The unary coefficient curve determines a canonical modular λ-coordinate. The project reconstructs a lift to the upper half-plane and an exact real-frame horocycle. On the level-two frame quotient, a nonperiodic horocycle is equidistributed for normalized Haar probability, whereas periodic horocycles are closed cusp orbits. The specialized Dani–Smillie theorem is proved through:

1. quantitative nonescape of occupation measures;
2. avoidance of mass on the countable periodic family;
3. good return times for compactly supported tests;
4. the nonergodic pointwise ergodic theorem; and
5. uniqueness of an invariant probability off the periodic set.

The final unconditional statements are `gammaTwoPointwiseHorocycleEquidistribution_unconditional` and `gammaTwoDaniSmillieTheorem_unconditional` in `HGAS/Modular/GammaTwoOffPeriodicUnique.lean`. The argument formalizes the level-two instance of Dani's invariant-probability classification [8] and the periodic-or-equidistributed orbit dichotomy of Dani and Smillie [9]; Hubbard and Miller provide a modern finite-area treatment of the latter [13].

The selected detector orbit cannot lie in the Haar-generic branch. A compactly supported family of theta tests has finite value along the selected compact orbit, but its Haar cusp moment diverges. Exact orientation reversal and constant time rescaling handle either sign of ω_j. Therefore the reconstructed frame orbit is periodic. The composition is `standardizedUnaryColumn_hasHorizontalOrPeriodicExactFrameLift_unconditional` in `HGAS/Arithmetic/NormalizedCorrectedFixedDetectorStandardizedExactFrameExclusion.lean`.

7.2 The finite closed-cusp selector

A periodic frame orbit admits a horizontal chart

$$\tau(t) = x + st + iy, \qquad y > 0, \quad s \neq 0. \tag{7.3}$$

The two compiled raw moments have Cesàro limit 1. Fourier series for $\vartheta_2, \vartheta_3, \vartheta_4$ on (7.3) show that the wrong cusp branches have zero mean. In the surviving infinity cusp, the means force the coherent root scales and quartic amplitude to equal 1. Thus, after the exact resampling back from (7.1),

$$Y(t) = \vartheta_3(\tau(t)), \qquad p_4(t) = \vartheta_4(\tau(t))^2. \tag{7.4}$$

This is proved in `HGAS/Modular/FiniteClosedCuspMeanSelector.lean`, `HGAS/Modular/FiniteClosedCuspResampledMeanSelector.lean`, and `HGAS/Arithmetic/NormalizedCorrectedFixedDetectorFiniteClosedBranchSelection.lean`.

The invariant residuals now recover every unary coefficient:

$$\begin{aligned} p_4^2 - C = 0 &\Longrightarrow C = \vartheta_4^4, \\ Y^4 - A - C = 0 &\Longrightarrow A = \vartheta_2^4, \\ 12H - E = 0 &\Longrightarrow H = E_2/12. \end{aligned} \tag{7.5}$$

For the last identity, the common nonvanishing $Y = \vartheta_3$ coordinate and its polynomial differential equation allow cancellation of the clock and Y, forcing equality of the two E-coordinates.

Only the sign of p_2 remains. Since

$$p_2^2 = \vartheta_2(\tau)^4,$$

we have $p_2 = \pm\vartheta_2(\tau)^2$. Translation $\tau \mapsto \tau + 2$ changes the sign of ϑ_2^2 while fixing the other canonical root coordinates. Hence one of τ and $\tau + 2$ is an exact canonical seed for (p_2, p_3, p_4, H). This argument is `exists_canonicalRootSeed_of_finiteClosedSelection` in `HGAS/Modular/FiniteClosedCuspCanonicalRootSeed.lean`.

7.3 Global affine Bloch continuation

At the canonical seed, the two Jacobi quadrics vanish. Along the complete trajectory, (6.3) prevents the projective coordinate T_1 from vanishing. Pointwise theta saturation reconstructs affine Bloch parameters, and uniqueness for the regular polynomial ODE propagates the identification for all real time. Thus the physical column is a globally regular affine Jacobi–Bloch trajectory.

The detector-facing result is `physicalColumn_exists_classifiedSeed` in `HGAS/Arithmetic/NormalizedCorrectedFixedDetectorPhysicalClassifiedSeed.lean`. It also chooses the actual initial phase and the unique free gauge constant matching the physical state at time zero. No canonical all-one phase is substituted.

8 The repaired three-moment theorem

This is the most delicate identification in the lower bound. We give it in detail, including the failed zero-ratio branch which motivated the DQL augmentation.

8.1 Quadratic–geometric coefficients

After multiplication by the fourth-root gauge, an affine Bloch theta trace has coefficients

$$C_k = C\, R^{k^2} S^k, \qquad k \in \mathbb{Z}, \tag{8.1}$$

with $R, S \neq 0$. Integer translations $k \mapsto k - \ell$ and reflected translations $k \mapsto -k - \ell$ retain the same form. The direct factorization is proved in `HGAS/Modular/AffineBlochQuadraticGeometric.lean`.

A normalized sequence $R^{k^2} S^k$ is determined by its coefficients at $k = 1$ and $k = -1$. Indeed their product determines R^2, so either $(R, S) = (R', S')$ or $(R, S) = (-R', -S')$. The second alternative gives the same sequence because $k^2 + k$ is even. This elementary rigidity lemma is formalized in `HGAS/Modular/QuadraticGeometricSequence.lean`.

8.2 What the three moments select

Let $\Omega \neq 0$ be the private angular frequency and put

$$\rho = \frac{N}{\Omega}. \tag{8.2}$$

The quadratic dispersion of the trace makes its mean resonance equation factor into two Bloch classes. If ρ is irrational, the two possible classes are disjoint:

$$a \in \mathbb{Z} \qquad \text{or} \qquad a \in -\rho + \mathbb{Z}. \tag{8.3}$$

In the first class, the zeroth and the two twisted first moments select

$$C_{-\ell}, \quad C_{1-\ell}, \quad C_{-1-\ell}. \tag{8.4}$$

In the reflected class they select the same three neighboring coefficients in reverse order. The normalized zeroth moment is 1; the $+1$ and -1 moments match the corresponding target theta coefficients. By the quadratic–geometric lemma, every coefficient matches the target, hence the whole theta trace matches.

The twisting constants are the actual initial clock characters. Their norm is one, so they are nonzero and may be cancelled; they need not equal one. Arbitrary recurrent horizon sequences are permitted. The exact phase-free theorem is `quadraticTrace_eq_target_of_three_moments` in `HGAS/Modular/JacobiThreeMomentTargetRigidity.lean`.

8.3 Why $\rho = 0$ cannot be ignored

Warning 8.1 (A genuine discarded argument)**.** *When $\rho = 0$, the direct and reflected resonance classes in (8.3) coincide. The three averages may then see sums of two coefficients rather than one coefficient, and the preceding proof is false.*

This is not merely a missing side condition. There are quadratic–geometric counterfamilies

$$C_k = \lambda^{k^2} A^{k^2} B^k, \qquad |A| = 1, \qquad B + B^{-1} = 2A^{-1}, \tag{8.5}$$

whose zeroth and two first resonant moments have the prescribed values although the trace is not the canonical target. For example, one may take

$$A = i, \qquad B = i(\sqrt{2} - 1). \tag{8.6}$$

Thus an older theorem which admitted the zero-ratio branch is not used in the final dependency chain.

The repair is exactly Section 3.4. Let ℓ_j be the nonzero linear DQL character of column j, and let e_j be its private character. The selected torus frequency Ω_{tor} has no nontrivial integer annihilator. If

$$\frac{\langle \ell_j, \Omega_{\mathrm{tor}} \rangle}{\langle e_j, \Omega_{\mathrm{tor}} \rangle} = \frac{a}{b} \in \mathbb{Q}, \tag{8.7}$$

then

$$b\ell_j - ae_j \tag{8.8}$$

annihilates Ω_{tor}. Nonresonance makes (8.8) the zero character. Its private coordinate gives $a = 0$; a nonzero equation coordinate of ℓ_j then gives $b = 0$, contradicting the choice of denominator. Therefore (8.7) is irrational for every augmented column.

This finite-character proof is `linear_div_private_pairing_irrational` and its physical-frequency specialization `numberFrequency_div_omegaFrequency_irrational` in `HGAS/Arithmetic/NormalizedCorrectedEffectiveColumnFrequencyIrrational.lean`.

8.4 Canonical theta output

Freeze the target Gaussian theta coefficients at the actual initial detector phase. They are quadratic–geometric, and the compiled $+1$ and -1 selector characters multiply the two neighboring target coefficients to $1/256$. The corrected moment limits and irrationality argument therefore give, for every column j and every $t \geq 0$,

$$\mathrm{gauge}_j(t)\, T_{1,j}(t) = \sum_{n\in\mathbb{Z}} \mathrm{thetaTerm}_{I,j,n}(1/256, \theta(t)). \tag{8.9}$$

This is exactly `hasCanonicalThetaOutputs` in `HGAS/Arithmetic/NormalizedCorrectedEffectiveCanonicalThetaOutputs.lean`. Its sole structural premise is that every linear column is nonzero, which (3.12) supplies for every final universal instance.

9 The yes-instance nonzero-mean theorem

Assume I is solvable and suppose, toward a contradiction, that every conditional compiled drift coordinate has mean zero. Section 6 then supplies a selected bidirectionally recurrent validity orbit O.

9.1 Exact factorization of the physical detector

The selected orbit begins at an arbitrary physical phase. Let $\theta(t) = \theta(0) + t\Omega$ be its detector phase in turns. The right-hand-side oscillator has the same frequency as the flattened DQL right-hand-side character, including the compiler's factor -4. Uniqueness for the corresponding linear complex ODE gives

$$R_{\mathrm{rhs}}(t) = \zeta\, \chi_{\mathrm{rhs}}(\theta(t)), \qquad \zeta \neq 0. \tag{9.1}$$

The constant ζ is retained; it is not silently normalized to one.

The reciprocal graph residuals give the two Poisson factors

$$G_+ = (1 - \tfrac{1}{2}W)^{-1}, \qquad G_- = (1 - \tfrac{1}{2}W^{-1})^{-1}, \tag{9.2}$$

where $|W| = 1$. The absolutely convergent two-sided geometric identity

$$\sum_{s\in\mathbb{Z}} \left(\frac{1}{2}\right)^{|s|} W^s = \frac{3/4}{(1 - \frac{1}{2}W)(1 - \frac{1}{2}W^{-1})} \tag{9.3}$$

identifies the compiler's literal factor $(3/4)G_+G_-$ with the desired Poisson series.

Substituting the canonical theta output (8.9), multiplying the finitely many column factors, and using absolute summability yields the pointwise identity

$$\mathrm{detectorValue}_I(O(t)) = \zeta\, \mathcal{D}_I(\theta(t)) \qquad (t \geq 0). \tag{9.4}$$

All phase, sign, fourth-power, 2π, $3/4$, and $1/256$ normalizations in (9.1)–(9.4) are proved in `HGAS/Arithmetic/NormalizedCorrectedFixedDetectorSelectedPhysicalPhase.lean` and `HGAS/Arithmetic/NormalizedCorrectedFixedDetectorSelectedFactorization.lean`.

9.2 Contradictory Cesàro limits

By (5.9), (5.10), and (9.4), along the selected physical horizons,

$$\frac{1}{T_N}\int_0^{T_N} \mathrm{detectorValue}_I(O(t))\ \mathrm{d}t \longrightarrow \zeta \int_{\mathbb{T}^{V+E}} \mathcal{D}_I\ \mathrm{d}\theta. \tag{9.5}$$

Because I is solvable, (5.5) makes the integral nonzero; because $\zeta \neq 0$, the limit in (9.5) is nonzero.

On the other hand, the selected orbit was obtained from the assumption that every conditional drift mean is zero. The detector drift coordinate therefore has Cesàro limit zero along the same physical horizons. A Hausdorff space has unique limits, giving a contradiction.

The analytic statement is `false_of_solvable_and_canonicalThetaOutputs` in `HGAS/Arithmetic/NormalizedCorrectedEffectiveSelectedDetectorAverage.lean`. The passage back to conditional invariant means is `not_conditionalBaseDriftZero_of_solvable` in `HGAS/Arithmetic/NormalizedCorrectedEffectiveNonzeroMeanFromTheta.lean`. Consequently: if μ is an ergodic invariant probability whose finite chart has positive mass, and μ_{fin} is its conditional probability there, then

$$I \text{ solvable} \quad \Longrightarrow \quad \exists q \qquad \int b_{I,q}\ \mathrm{d}\mu_{\mathrm{fin}} \neq 0. \tag{9.6}$$

10 The no-instance invariant finite graph

Now assume I is not solvable. The construction must produce one invariant probability with strictly positive reward. This branch does not use the selected-orbit contradiction from Section 9.

10.1 The orbit-closure Fourier model

Choose the explicit positive unit equilibrium of every prime controller. Its phase flow is a translation with frequencies

$$\frac{1}{2\pi}\left(\varepsilon_\ell \sqrt{p_\ell}\right)_\ell. \tag{10.1}$$

The closure of its orbit is a compact torus, and the orbit is dense there. Every compiled drift coordinate restricts to an absolutely convergent real Fourier family

$$g_q(\theta) = \sum_{r \in \mathcal{I}_q} c_{q,r}\, e^{2\pi i \langle k_{q,r}, \theta \rangle}. \tag{10.2}$$

The complete finite-code-to-Fourier crosswalk is in `HGAS/Analysis/NormalizedCorrectedCanonicalDriftFourierFamily.lean` and `HGAS/Arithmetic/NormalizedCorrectedCanonicalOrbitClosureModel.lean`.

When I has no solution, every zero Fourier mode in (10.2) vanishes. For the detector coordinate this is precisely the contrapositive of the constant-term calculation (5.4); the other validity and centered-moment coordinates have zero modes by their explicit normalization. Thus

$$c_{q,r} = 0 \quad \text{whenever} \quad k_{q,r} = 0. \tag{10.3}$$

10.2 Small divisors and a bounded primitive

For a nonzero integer character k, its frequency

$$\langle k, \Omega \rangle = \frac{1}{2\pi} \sum_{\ell} k_\ell \varepsilon_\ell \sqrt{p_\ell} \tag{10.4}$$

is nonzero. More is needed: division by (10.4) must preserve convergence. The sum in (10.4) is an algebraic number in the finite multiquadratic extension generated by the selected square roots. Taking the product of all Galois conjugates gives a nonzero rational integer after denominators are cleared. Bounding all conjugate factors except the original one gives an effective lower bound for $|\langle k, \Omega \rangle|$ in terms of the radical conjugate height of k.

The detector coefficients have Gaussian/geometric decay. The development proves the weighted summability

$$\sum_{r} |c_{q,r}| \, H(k_{q,r})^{D_{\mathrm{rad}}-1} < \infty, \tag{10.5}$$

which dominates the reciprocal small divisor. Therefore

$$U_q(\theta) = \sum_{r : k_{q,r} \neq 0} \frac{c_{q,r}}{2\pi i \langle k_{q,r}, \Omega \rangle} e^{2\pi i \langle k_{q,r}, \theta \rangle} \tag{10.6}$$

converges uniformly and differentiates along the translation flow to g_q. The radical lower bound, summability budget, and primitive construction occur in `HGAS/Clocks/RadicalSmallDivisor.lean`, `HGAS/Analysis/CanonicalEffectivePrimitiveBudget.lean`, and `HGAS/Analysis/NormalizedCorrectedCanonicalEffectivePrimitiveBudget.lean`.

10.3 Embedding the primitive as an invariant graph

Scale the primitive vector by a positive rational gain κ. The gain is chosen effectively from exact finite coefficient bounds so that the graph lies well inside the finite stereographic accumulator chart. The base modular/Jacobi orbit is simultaneously centrally compactified. There is a fixed constant $c_{\mathrm{cen}} > 0$ such that the product W_0 of squared central chart weights satisfies

$$W_0 \geq c_{\mathrm{cen}} \tag{10.7}$$

on the selected orbit closure. The gain is small enough that the stereographic scalar coordinate W of the primitive graph satisfies

$$W \geq 1 - \frac{c_{\mathrm{cen}}}{4}. \tag{10.8}$$

The source reward is

$$\mathcal{R} = W + 2W_0 - 1 - \varepsilon. \tag{10.9}$$

The effective reward margin satisfies $\varepsilon \leq 3c_{\mathrm{cen}}/8$. The formal estimates give

$$\mathcal{R} \geq \delta := \frac{3c_{\mathrm{cen}}}{8} > 0 \tag{10.10}$$

on the invariant graph. Pushing Haar probability on the orbit-closure torus to this graph produces the required invariant probability.

The exact endpoint is `finite_graph` in `HGAS/Arithmetic/NormalizedCorrectedEffectiveNoInstanceFiniteGraph.lean`; its explicit margin is proved in `HGAS/Arithmetic/NormalizedCorrectedFixedDetectorExplicitFiniteGraphGap.lean`.

11 Compactification and the strict sign dichotomy

11.1 Central compactification

Affine source blocks are mapped to central spheres. If $y = x/w$ is an affine chart and $P(y)$ is polynomial, multiplying by a sufficiently large common even power of the central weights clears every denominator. The resulting polynomial field is tangent to every sphere factor, including points on the equator. Inside the finite chart it is a positive time change of the original source. The sparse rational compiler and its global tangency theorem are in `HGAS/Compactification/MixedCentralPolynomialCode.lean`, `HGAS/Compactification/MixedCentralPolynomialField.lean`, and `HGAS/Compactification/MixedCentralPolynomialGlobalTangency.lean`.

The final normalized source uses two sphere blocks: the centrally compactified base and the accumulator sphere. Degree bounds and the common clearing exponent are computed from finite syntax and are constant throughout the fixed universal family.

11.2 The stereographic accumulator

For a drift vector $g = (g_1, \dots, g_m)$, adjoin a sphere coordinate $(U, W) \in \mathbb{R}^m \times \mathbb{R}$, $\|U\|^2 + W^2 = 1$, with equations

$$\begin{aligned} U_i' &= (1 + W)g_i - (U \cdot g)U_i, \\ W' &= -(1 + W)(U \cdot g). \end{aligned} \tag{11.1}$$

These are the inverse-stereographic compactification of translation by the primitive of g. Direct differentiation proves tangency to the accumulator sphere. The literal polynomial code and evaluation theorem are in `HGAS/Accumulator/PolynomialCode.lean`.

If an invariant probability gives positive mass to the finite chart $W \neq -1$, condition it on that invariant chart and integrate the derivative of the inverse stereographic coordinates. Invariance makes each derivative integral zero; the chart equations reduce this to

$$\int g_i \, \mathrm{d}\mu = 0 \quad \text{for every } i. \tag{11.2}$$

This compact drift obstruction is proved in `HGAS/Accumulator/CompactDriftObstruction.lean`.

11.3 Reward on boundary and infinity

Recall (10.9). On a central boundary face, $W_0 = 0$, and $W \leq 1$, so

$$\mathcal{R} \leq -\varepsilon. \tag{11.3}$$

At accumulator infinity, $W = -1$ and $W_0 \leq 1$, giving the same bound. The elementary inequalities are in `HGAS/Accumulator/RewardGap.lean`.

11.4 The two cases

Let P be the proposition "I is solvable" and choose an ergodic invariant probability maximizing $\int \mathcal{R}$.

If P holds, ergodicity makes the invariant finite chart have measure zero or one. If it had measure one, (11.2) would say that every conditional drift mean is zero, contradicting (9.6). It therefore has measure zero, and (11.3) gives

$$\mathcal{M}(\mathcal{R}) \leq -\varepsilon < 0. \tag{11.4}$$

If P fails, Section 10 supplies an invariant graph probability with reward average at least $\delta > 0$, so

$$\mathcal{M}(\mathcal{R}) \geq \delta > 0. \tag{11.5}$$

Together,

$$\boxed{\mathcal{M}(\mathcal{R}) < 0 \quad \Longleftrightarrow \quad I \text{ is solvable.}} \tag{11.6}$$

This abstract two-case proof is `maximalInvariantAverage_neg_iff_of_compactAccumulatorGap` in `HGAS/Accumulator/CompactSignDichotomy.lean`.

The two instance-dependent facts (9.6) and (10.10) are packaged by `gapOfCanonicalThetaOutputs` in `HGAS/Arithmetic/NormalizedCorrectedEffectiveGapFromCanonicalTheta.lean`. For the augmented universal family, the resulting unconditional gap is `gap` in `HGAS/Arithmetic/NormalizedCorrectedAugmentedEffectiveFamily.lean`.

12 The homogeneous product-sphere wrapper

At this stage the source is a rational polynomial flow tangent to a compact product of spheres, with reward $\mathcal{R}$, and (11.6) is known. The wrapper converts this compact angular system into one homogeneous polynomial field on an ordinary Euclidean space.

12.1 A rational two-sheeted chart

Add one coordinate z. On the Euclidean unit sphere, use the two sheets

$$(k+1)z^2 = 1, \tag{12.1}$$

where k is the number of source sphere blocks. Source coordinates are represented as zy. If the source field/reward degree bound is the odd integer D, multiplying chart polynomials by the rational integer $(k+1)^{(D-1)/2}$ cancels the power of z on both sheets. No $\sqrt{k+1}$ coefficient is introduced.

Scaled block radii

$$s_b = (k+1)\sum_{j \in b} x_j^2 \tag{12.2}$$

equal 1 on the intended full stratum. Division-free tangentialization makes the angular code tangent to every block sphere. A radius-cleanup field drives every other block-radius configuration toward the full stratum while leaving proper faces invariant. Proper faces receive reward -1; on the full stratum the reward is exactly the source reward $\mathcal{R}$.

12.2 Parity homogenization

The raw angular and reward polynomials have the correct parity but may contain several lower degrees. Multiplying each monomial by a suitable power of $\|x\|^2$ makes the angular part homogeneous of

$$d = D + 2(k+1) \tag{12.3}$$

and the reward homogeneous of degree $d-1$, without changing either function on the unit sphere. The final field is

$$F_{\mathrm{fin}}(x) = Y_{\mathrm{hom}}(x) + A_{\mathrm{hom}}(x)x. \tag{12.4}$$

It is rational and homogeneous of odd degree d, and on the sphere its angular field and radial reward are exactly the wrapped source data.

Invariant-average bookkeeping on the two sheets and proper faces proves

$$0 \text{ is GAS for } F_{\mathrm{fin}} \quad \Longleftrightarrow \quad \mathcal{M}(\mathcal{R}) < 0. \tag{12.5}$$

The finite construction is `HGAS/Wrapper/UnitSpherePolynomialWrapper.lean`; the exact GAS equivalence is `CompactSignSourceCertificate.finalField_gas_iff_solvable` in `HGAS/Compiler/CompactSignSourceWrapper.lean`.

Combining (11.6) and (12.5) gives, for every augmented DQL instance,

$$0 \text{ is GAS for the emitted homogeneous field} \quad \Longleftrightarrow \quad I \text{ is solvable.} \tag{12.6}$$

13 Freezing the resources and proving hardness

The remaining question is effectivity. A semantic equivalence such as (12.6) is insufficient unless the finite output code is computable from the universal-halting input and its dimension and degree are independent of that input.

13.1 Why the resource bounds are fixed

In the fixed universal DQL family, input specialization changes right-hand-side constants but not the quadratic and linear character tables. Augmentation preserves this common character data. Every compiler degree—the theta detector degree, central clearing exponent, accumulator degree, wrapper degree, and final homogeneous degree—is a function of the fixed row/variable counts and common character table. Therefore it is independent of t.

This is formalized by the `SameCharacterData` lemmas in `HGAS/Arithmetic/NormalizedCorrectedAugmentedUniversalResources.lean`. The same file fixes:

- the compact source dimension;
- the two-block sphere assignment;
- a proof that the source product sphere is nonempty;
- the common odd wrapper degree; and
- the computable finite source field/reward code.

13.2 The effectivity firewall

The structure

`FixedUniversalDQLCompactSignSourceFamily`

in `HGAS/Compiler/FixedUniversalDQLCompactSignSourceFamily.lean` separates finite code from semantic proof data. Its parameters and fields supply:

1. exact DQL solvability versus unary universal halting;
2. one fixed source block assignment, a nonempty source product sphere, and one fixed odd wrapper degree at least three;
3. a compact-sign certificate for every input;
4. one raw output-code function;
5. equality of that raw code with the certificate's field; and
6. computability of the raw output-code function.

The certificate may contain flows, invariant probabilities, real limits, and classical choices, but the emitted code function is built only from finite rational syntax. Its computability theorem composes primitive- recursive DQL augmentation, controller parameter generation, complex realification, polynomial compactification, accumulator code, and the homogeneous wrapper.

For the final family, the theorem

`hgasQ_output_iff_predicate`

states, for every t,

$$\mathrm{HGAS}_{\mathbb{Q}}(n_*, d_*)(\mathrm{outputCode}(t)) \quad \Longleftrightarrow \quad \mathrm{Predicate}(t). \tag{13.1}$$

Since outputCode is computable, (13.1) is a computable many-one reduction. The c.e. upper bound from Theorem 2.1 then proves

$$\mathrm{HGAS}_{\mathbb{Q}}(n_*, d_*) \text{ is } \Sigma^0_1\text{-complete.} \tag{13.2}$$

This is precisely `fixedPair_sigmaOneComplete`.

14 Cubicization and cofinal propagation

Write $d_* = 2k_* + 1$, with $k_* \geq 1$. There are two fixed-parameter computable finite-code transformations.

14.1 The stabilized Veronese cubic lift

Map $x \in \mathbb{R}^{n_*}$ to the vector $v_{k_*}(x)$ of all monomials of degree k_*. If F has degree $2k_* + 1$, differentiating $v_{k_*}(x(t))$ yields a cubic polynomial expression in the Veronese coordinates on the Veronese variety. Away from that variety, add a fixed rational cubic penalty which drives the toric defining relations to zero.

The formal construction fixes one natural multiplier $M_{n,k}$ for each fixed pair (n,k), using `Classical.choice` on the proved geometric existence theorem. For an input field F, the emitted gain is

$$M_{n,k}(1+\|F\|_{1,\mathrm{coeff}}).$$

The coefficient norm is executable, and `Primrec.const` $M_{n,k}$ makes the resulting map primitive recursive in F. The formalization therefore proves fixed-(n,k) computability; it does not formalize real quantifier elimination, numerically extract $M_{n,k}$, or give one algorithm computing the multiplier from varying (n,k). The manuscript gives a paper-level uniform strengthening by terminating rational searches with real quantifier elimination.

The penalty creates a uniformly negative transverse angular block, while on the toric core the radial cocycle is the source cocycle. Splitting invariant probabilities between the core and its complement proves the exact equivalence

$$F \text{ is GAS} \quad \Longleftrightarrow \quad \mathrm{CubicLift}(F) \text{ is GAS.} \tag{14.1}$$

Malformed source codes are sent to a syntactically homogeneous cubic negative instance, so the map is correct on all raw inputs.

The semantic theorem is `sourceGAS_iff_stabilizedCubicGAS` in `HGAS/Veronese/CubicLiftGASConjugacy.lean`. The primitive- recursive sparse compiler and fixed-parameter gain are in `HGAS/Computability/CubicLiftCodeCompiler.lean` and `HGAS/Computability/CubicLiftEffectiveGain.lean`. Its target dimension is exactly N_* from (1.4).

14.2 Dimension and odd-degree padding

From a cubic field, adjoin any fixed number of independent cubic sinks. To raise degree from 3 to $3+2j$, multiply the old block by a suitable even radial power and give the new sink coordinates compatible homogeneous negative dynamics. Product-GAS and homogeneity calculations give exact preservation and reflection of GAS. The finite compiler is `HGAS/Propagation/DegreePaddingCode.lean`.

For $N \geq N_*$, take $N - N_*$ sink dimensions. Every odd $d \geq 3$ has the form

$$d = 3 + 2\frac{d-3}{2}. \tag{14.2}$$

Thus the fixed-pair reduction (13.1) propagates to every target pair in Theorem 1.2. The exact composition is `hgasQ_manyOneReducible_cofinalFixedPair` in `HGAS/Propagation/CofinalFixedPairReduction.lean`, and Σ^0_1-membership is supplied again by Theorem 2.1. This completes the proof of Theorem 1.2.

15 Relation to the literature

15.1 Stability and decidability

Classical work on homogeneous systems establishes strong structural properties and converse Lyapunov theorems; see Samardžija [22] and Rosier [21]. Ahmadi proved strong NP-hardness for

deciding asymptotic stability of cubic homogeneous vector fields and explicitly isolated the decidability question [1]. Ahmadi and El Khadir proved algebraic/rational Lyapunov results for stable homogeneous polynomial fields [2]; Ahmadi and Parrilo studied SOS certificates in several structured settings [3].

The result here is of a different recursion-theoretic strength. NP-hardness does not rule out an algorithm; Σ_1^0-completeness does. Moreover, the lower bound already holds for one fixed dimension and degree, and after the Veronese lift it holds at cubic degree in every sufficiently large fixed dimension. The c.e. upper bound identifies the exact arithmetical level and shows that positive instances can be certified even though no total decision procedure exists.

There is a broader literature on undecidability and computational universality for polynomial ODEs, for example Buser and Scarpellini [6] and Graça, Campagnolo, and Buescu [12]. Those results motivate the general question but do not directly give the fixed homogeneous-GAS statement: homogeneity, a single equilibrium at the cone tip, forward completeness, and global attraction impose a substantially different geometry.

15.2 Arithmetic and special-function inputs

The arithmetic ancestry is the DPRM theorem [10, 16, 17, 18]. The formal project proves the required primitive- recursive Diophantine closure internally, then uses the elementary four-square and multiplication-gate reductions to reach DQL form.

The finite polynomial theta system follows identities standard in the theory of Jacobi theta functions; useful references are Chandrasekharan [7], Mumford [20], and the NIST DLMF [11]. Polynomial dynamical formulations closely related to the one used here appear in [5, 4]. The horocycle branch uses the level-two specialization of the classical periodic-or- equidistributed dichotomy of Dani–Smillie [9, 13].

These citations identify provenance and comparison. They are not opaque proof constants in the Lean theorem. The concrete theta identities, finite quotient, nonescape estimates, occupation-measure argument, pointwise equidistribution statement, and every normalization used by the compiler have formal proofs in the dependency closure.

Kernel checking establishes the stated formal implication; it does not establish bibliographic priority or replace specialist peer review. This is a new manuscript-level result and should be treated as unrefereed until the reduction and its relationship to the broader literature have received independent expert review.

15.3 Scope of the present result

The formal theorem settles the general and fixed-resource decidability question for rational homogeneous polynomial fields in the stated cofinal range. It leaves at least three natural quantitative problems:

1. evaluate or drastically reduce the chosen fixed constants;
2. determine the exact low-dimensional boundary, especially $\mathrm{HGAS}_{\mathbb{Q}}(3,3)$; and
3. find a substantially simpler reduction which avoids the modular source while preserving fixed cubic resources.

The first two are not consequences of the present proof. In particular, the phrase “cubic undecidability” throughout this document always means cubic degree in every sufficiently large fixed dimension.

16 Lean correspondence and reusable modules

The formalization was organized so that the manuscript proof and code have the same narrow interfaces. The following table lists the principal crosswalk.

Mathematical step	Principal module or theorem
Raw rational field language	`HGAS/Computability/HGASLanguage.lean`; `HGASQ`
Σ_1^0 vocabulary and universal source	`HGAS/Computability/SigmaOne.lean`; `universalHalts_sigmaOneComplete`
C.e. upper bound	`HGAS/Lyapunov/RationalHomogeneousConverse.lean`; `hgasQ_re_of_oddDegree`
Primitive-recursive Diophantine closure	`HGAS/Arithmetic/PrimitiveRecursiveDiophantine.lean`; `binaryPrimrecRel_dioph`
Unconditional fixed DQL	`HGAS/Arithmetic/UnconditionalMRDP.lean` and `HGAS/Arithmetic/UnconditionalFixedDQL.lean`
Nonzero-column repair	`HGAS/Arithmetic/DQLNonzeroLinearAugmentation.lean`; `augment_solvable_iff`
Positive Fourier selector	`HGAS/Arithmetic/PositiveConstantTerm.lean` and `HGAS/Arithmetic/DetectorFourier.lean`
Prime-radical nonresonance	`HGAS/Clocks/RadicalIndependence.lean` and `HGAS/Arithmetic/NormalizedCorrectedEffectiveSelectedDetectorPhase.lean`; `physicalDetectorTurnsFrequency_hasNoCharacter`
Polynomial Jacobi compiler	`HGAS/Modular/JacobiPolynomialCode.lean` and `HGAS/Modular/LinkedJacobiPolynomialCode.lean`
Bidirectional recurrent selection	`HGAS/Arithmetic/NormalizedCorrectedFixedDetectorBidirectionalValidityOrbit.lean`
Level-two equidistribution	`HGAS/Modular/GammaTwoOffPeriodicUnique.lean`; `gammaTwoDaniSmillieTheorem_unconditional`
Canonical root and Bloch continuation	`HGAS/Arithmetic/NormalizedCorrectedFixedDetectorCanonicalRootSeed.lean` and `HGAS/Arithmetic/NormalizedCorrectedFixedDetectorPhysicalClassifiedSeed.lean`
Repaired three-moment rigidity	`HGAS/Modular/JacobiThreeMomentTargetRigidity.lean` and `HGAS/Arithmetic/NormalizedCorrectedEffectiveColumnFrequencyIrrational.lean`
Canonical theta output	`HGAS/Arithmetic/NormalizedCorrectedEffectiveCanonicalThetaOutputs.lean`
Yes-instance nonzero drift	`HGAS/Arithmetic/NormalizedCorrectedEffectiveNonzeroMeanFromTheta.lean`
No-instance finite graph	`HGAS/Arithmetic/NormalizedCorrectedEffectiveNoInstanceFiniteGraph.lean`
Compact sign dichotomy	`HGAS/Accumulator/CompactSignDichotomy.lean` and `HGAS/Arithmetic/NormalizedCorrectedEffectiveGapFromCanonicalTheta.lean`
Homogeneous wrapper	`HGAS/Compiler/CompactSignSourceWrapper.lean` and `HGAS/Wrapper/UnitSpherePolynomialWrapper.lean`
Fixed effective family	`HGAS/Compiler/FixedUniversalDQLCompactSignSourceFamily.lean` and `HGAS/Arithmetic/NormalizedCorrectedAugmentedEffectiveFamily.lean`
Cubic/cofinal propagation	`HGAS/Computability/CubicLiftEffectiveGain.lean` and `HGAS/Propagation/CofinalFixedPairReduction.lean` and `HGAS/Propagation/CofinalSigmaOne.lean`

Several abstractions are intentionally more reusable than this application:

- `DQLNonzeroLinearAugmentation.augment` applies to every finite DQL system.
- `FixedUniversalDQLCompactSignSourceFamily` accepts any fixed-arity DQL family with the

required semantic and computable-code fields.

- `quadraticTrace_eq_target_of_three_moments` is independent of theta functions. Its reusable interface takes quadratic–geometric source and target data, nonzero private frequency and twists, an irrational ratio, summability, divergent horizons, and three moment limits.
- the invariant-average, ergodic-maximizer, compact-accumulator, and product-sphere wrapper theorems do not mention universal halting.
- the stabilized Veronese GAS reduction accepts every finite homogeneous field code in positive dimension and odd degree at least three.

17 Trust, audits, and reproducibility

17.1 No hidden mathematical postulate

The final theorem imports `HGAS/Arithmetic/NormalizedCorrectedAugmentedEffectiveFamily.lean` through `HGAS.lean`. The source tree contains no declaration matching a strict scan for project `axiom`, `unsafe`, `sorry`, or `admit`. In particular:

- MRDP closure is a theorem, not an axiom;
- the level-two Dani–Smillie statement is a theorem, not an interface hypothesis;
- canonical theta outputs are proved for the augmented family;
- the no-instance graph is constructed, not assumed; and
- the final family has no remaining gap or theta parameter.

Running `#print axioms` on `fixedPair_sigmaOneComplete`, `cofinal_sigmaOneComplete`, `cubic_sigmaOneComplete`, `fixedPair_not_computable`, and `cubic_not_computable` (as well as the two named non-c.e.-complement consequences) returns exactly

$$[\texttt{propext},\ \texttt{Classical.choice},\ \texttt{Quot.sound}]. \tag{16.1}$$

These are ordinary logical foundations of Lean/Mathlib; there is no project-specific axiom and no `sorryAx`.

The audited environment is Lean 4.30.0 with Mathlib tag $v4.30.0$, commit `c5ea00351c28e24afc9f0f84379aa41082`. For the prover and library, see de Moura and Ullrich [14] and the Mathlib community [19].

17.2 Mechanical checks

From the `lean/` directory, the decisive commands are:

```
lake build HGAS

lake env lean -DwarningAsError=true \
  HGAS/Arithmetic/NormalizedCorrectedAugmentedEffectiveFamily.lean
```

```
rg -n \
  -e '^[[:space:]]*((private|protected)[[:space:]]+)?axiom[[:space:]]' \
  -e '^[[:space:]]*((private|protected)[[:space:]]+)?unsafe[[:space:]]' \
  -e '^[[:space:]]*(sorry|admit)\b' \
  -e ':=[[:space:]]*(by[[:space:]]+)?(sorry|admit)\b' \
  HGAS HGAS.lean
```

The public/default `HGAS` target build completes successfully; the endpoint file also recompiles successfully with warnings treated as errors, and the strict token scan is empty. Its transitive import closure contains 783 of the 824 modules under `HGAS/`. All 41 modules outside that closure were also elaborated directly with `-DwarningAsError=true`; all succeeded. Their exact list and legacy/audit status are recorded in `LEGACY_MODULES.md`. Independent audit reports are stored under `../aud its/` from this working directory. They separately check the MRDP chain, theta identities, clock normalization, quotient/horocycle arguments, selected phase factorization, compact accumulator, wrapper, cubic lift, augmented fixed resources, and the repaired endpoint.

17.3 Why classical choice does not spoil the reduction

The fixed MRDP polynomial and some analytic witnesses, including the geometry-only cubic gain multiplier for each fixed (n, k), are obtained by classical choice. This is compatible with a fixed-parameter computable many-one reduction because each chosen finite object is thereafter a constant. A program may contain any fixed finite lookup table. This does not assert a uniform extractor for those constants as the parameters vary. More importantly, the final family structure does not derive its output code by inspecting semantic proof objects: it stores a separate finite source-code function and a proof that this function is computable. The equality with the semantic certificate is proved afterward. This is the formal effectivity firewall described in Section 13.

17.4 Final logical compression

After all mathematics has been packaged, the endpoint proof is short:

$$\begin{aligned} \mathrm{Predicate}(t) &\iff \widetilde{I}_t \text{ solvable} \\ &\iff \mathcal{M}(\mathcal{R}_t) < 0 \\ &\iff \mathrm{outputCode}(t) \in \mathrm{HGAS}_{\mathbb{Q}}(n_*, d_*). \end{aligned} \tag{16.2}$$

The first equivalence is arithmetic and augmentation; the second is the two-field semantic gap; the third is the homogeneous wrapper. Computability of outputCode makes (16.2) a many-one reduction. Combining it with the c.e. upper bound proves fixed-pair Σ^0_1-completeness; the effective Veronese and padding maps give the cofinal theorem.

18 Conclusion

For rational homogeneous polynomial vector fields, the general GAS decision problem is not merely difficult: it is undecidable already at one fixed dimension and one fixed odd degree. Positive instances remain computably enumerable through exact rational Lyapunov certificates, so the

fixed language is precisely Σ_1^0-complete. A stabilized Veronese lift lowers the degree to three at the cost of a fixed increase in dimension, and padding propagates the result to every sufficiently large fixed dimension and every odd degree at least three.

The proof's delicate point is also its most instructive correction. Three moments do not determine a quadratic theta trace when the linear/private frequency ratio is zero. A tiny arithmetic preprocessing step, $x_j - y_j = 0$, both preserves solvability and makes the ratio irrational through prime-radical character independence. The repaired theorem is therefore stronger logically and cleaner modularly: every final universal instance passes through one uniform, reusable irrational-ratio selector.

The low-dimensional cubic case $(n, d) = (3, 3)$ remains open in this development. Settling it would be a separate result rather than a missing step in the theorem proved here.

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